

Date	30 September 2026	Team ID	SWTID-2026-26
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer	Maximum Marks	4 Marks

PROJECT DEVELOPMENT PHASE

Model Performance Test

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	• Source Schema: ServiceNow IT Service Management transactional tables (<code>sc_request</code> , <code>sc_req_item</code> , <code>sc_task</code> , <code>sysapproval_approver</code>). • Dataset Volume: 12,450 total procurement lifecycle records loaded into Power BI in-memory storage engine. • Star Schema Architecture: Fact table (<code>Fact_LaptopProcurement</code>) linked via 1-to-many relationships to <code>Dim_Date</code> , <code>Dim_Department</code> , <code>Dim_FulfillmentGroup</code> , and <code>Dim_ApprovalState</code> . • Rendered Data Fields: Request ID, RITM Number, SCTASK Number, Item Name (Lenovo ThinkPad X1 Carbon), Requester Department, Assignment Group (Hardware), Approval Status, and Unit Price (\$1,100).
2.	Data Preprocessing	• Missing Data Imputation: Replaced null closure values for in-progress tasks with dynamic platform execution timestamp. • Calculated Duration Metrics: Created custom duration columns: <code>Approval_Latency_Hours</code> (<code>sys_updated_on</code> - <code>sys_created_on</code>) and <code>Staging_Lead_Time_Hours</code> (<code>closed_at</code> - <code>sys_created_on</code>). • Type Transformations: Converted UTC timestamp strings to system datetime format and parsed raw currency strings to standard decimal types. • Anomaly & Sandbox Pruning: Filtered out test orders (<code>REQ0000001</code> to <code>REQ0000010</code>) and flagged automated records with <code>Automated_By_Flow</code> = <code>TRUE()</code> .
3.	Utilization of Data Filters	• Canvas Slicers: 1. <i>Date Hierarchy Slider:</i> Year, Quarter, Month, and dynamic sliding date window. 2. <i>Assignment Group Filter:</i> Pre-filtered to <code>Hardware</code> team queue with comparative peer visibility. 3. <i>Approval State Slicer:</i> Filter by <code>Approved</code> , <code>Pending</code> , and <code>Rejected</code> states. 4. <i>Item Configuration Slicer:</i> Multi-select for standard hardware specifications. • Page-Level Filters: Enforced <code>state != 7</code> (excluding cancelled requisitions) and filtered catalog items to <code>Standard Laptop</code> .
4.	DAX Queries Used	• Total Orders: <code>Total_Orders = COUNTROWS('Fact_LaptopProcurement')</code> • Average Staging Time (Hours): <code>Avg_Staging_Hours = CALCULATE(AVERAGE('Fact_LaptopProcurement'[Staging_Lead_Time_Hours]), 'Fact_LaptopProcurement'[State] = "Closed Complete")</code> • Flow Automation Compliance Rate: <code>Automation_Rate_% = DIVIDE(CALCULATE(COUNTROWS('Fact_LaptopProcurement'), 'Fact_LaptopProcurement'[Automated_By_Flow] = TRUE()), [Total_Orders], 0) * 100</code> • SLA Compliance (≤ 24h Staging): <code>SLA_Compliance_% = DIVIDE(CALCULATE([Total_Orders], 'Fact_LaptopProcurement'[Staging_Lead_Time_Hours] <= 24), [Total_Orders], 0) * 100</code>
5.	Dashboard design	No of Visualizations / Graphs: 6 Visualizations • 1. Executive KPI Cards: Total Orders (1,248) Avg Lead Time (4.2h) Automation Rate (100%) SLA (98.4%). • 2. Clustered Column & Line Chart: Monthly Order Requisition Volume vs. Staging Delivery Turnaround. • 3. Bar Chart: Configuration & Imaging Volume per Hardware Technician. • 4. Donut Chart: Requisitions by Requesting Department (Engineering, IT, Operations, Sales). • 5. Radial Gauge: SLA Fulfillment Rate against the 95% departmental service threshold. • 6. Process Funnel Chart: Stage Conversion (Request Submitted $\rightarrow$ Approved $\rightarrow$ Flow Triggered $\rightarrow$ Task Auto-Created $\rightarrow$ Delivered). 1,248 TOTAL ORDERS 4.2h AVG LEAD TIME 100% FLOW AUTOMATION 98.4% SLA COMPLIANCE
6.	Report Design	No of Visualizations / Graphs: 8 Visualizations • 1. Detail Drilldown Matrix: Itemized ledger (REQ, RITM, SCTASK, Requester, Technician, Created, Approved, Closed). • 2. Staging Heatmap Matrix: Turnaround latency by Day of Week vs. Hour of Order Submission. • 3. Waterfall Chart: Cumulative lead-time breakdown across approval, flow trigger, and physical staging. • 4. 100% Stacked Column Chart: Monthly requisition approval status ratios. • 5. Decomposition Tree: Root-cause analysis identifying delay drivers by technician shift. • 6. Ribbon Chart: Departmental laptop demand rankings quarter-over-quarter. • 7. Scatter Plot: Laptop Order Value vs. Fulfillment Time identifying queue anomalies. • 8. Audit Grid: Compliance audit log exportable for asset tracking in CMDB.

